Writing a Lab Report with LaTeX

Objective. The 5 to 6 lab sessions in this course will contribute to **a single report**, progressively completed throughout the semester. Each lab session should form an independent section (or chapter if you use the `report` class). The document should remain consistent, readable, and easy to maintain.

1. Organize files instead of writing everything in `main.tex`

```
report/
|-- main.tex
|-- src/
|   |-- packages.tex
|   |-- style.tex
|   |-- introduction.tex
|   |-- lab01.tex
|   |-- lab02.tex
|   `-- conclusion.tex
|-- figures/
|   |-- lab01/
|   `-- lab02/
`-- references.bib
```

```latex
\documentclass[a4paper,10pt]{article}
\input{src/packages.tex}
\input{src/style.tex}

\begin{document}
\maketitle
\tableofcontents
\input{src/introduction.tex}
\input{src/lab01.tex}
\input{src/lab02.tex}
% Add the remaining labs here.
\input{src/conclusion.tex}
\end{document}
```

The `main.tex` file controls the document: **structure and order only**. Place packages in `packages.tex`, formatting settings in `style.tex`, and the content of each lab in a separate file. The `\input{...}` command inserts the corresponding content. For example, `lab01.tex` should begin with `\section{Lab 1 - ...}` (without a new preamble).

2. Pay attention to typography and writing

- Remove automatic paragraph indentation with `\setlength{\parindent}{0pt}` and add a small paragraph spacing (`\parskip`). These settings belong in `style.tex`.
- Use `\section`, `\subsection`, etc., rather than manually formatting headings in bold. Keep the layout clean without excessive forced line breaks.
- For each lab, briefly present **the objective, method, results, and their interpretation**. A report should not be a simple collection of screenshots or code snippets.

3. Figures: readability, captions, and interpretation

- Prefer **vector graphics (PDF)** for plots and diagrams. For raster images (PNG), check resolution, cropping, text readability, and avoid excessive compression.
- Every figure must be **referenced in the text** (`Figure \ref{fig:lab1}`) and have an informative caption explaining **what is shown, under which conditions, and what can be concluded**. Avoid generic captions such as “Results” or “Screenshot”.

```latex
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.75\linewidth]{figures/lab01/time.pdf}
  \caption{Execution time as a function of input size for two methods.
  Method B becomes faster for inputs larger than 10^5 elements.}
  \label{fig:lab1}
\end{figure}
```

Key takeaway: a good report is more than a document that compiles successfully. Its structure should make it easy to read, and every result should be **explained and discussed**.